Mobile robot path planning based on improved A* algorithm

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***Abstract*—To solve the problems of A* algorithm, such as excessive number of redundant nodes, slow search time and sharp turn of path. a new approach is proposed: a five-neighborhood search A* algorithm that incorporates dynamic weighting in the heuristic function and utilizes second-order Bezier curve for path smoothing. Through the simulation environment built by PyCharm, it can be seen that the number of redundant nodes and the search time of the A* algorithm after dynamic weighting of five neighborhoods are reduced by 86.57% and 82.07% compared with the traditional eight-neighborhood search algorithm, and the cost remains the same in the precision of ten thousand bits. the path obtained after Bezier curve processing has no large angle turning points. After smooth optimization, the continuity of speed and acceleration can be maintained during the driving process, and the driving fluency can be improved.**

***Keywords*—A* algorithm; dynamic weighting; second-order Bezier curve; redundant nodes; search time**

I. Introduction

Path planning is an important part of mobile navigation technology of robots, reflecting the ability of robots to interact with the environment in the process of movement. An accurate path planning is the security guarantee for mobile robots to perform tasks. Researchers have given many effective algorithms in path planning, such as Simulated annealing algorithm, whale algorithm, A* algorithm, DWA algorithm, artificial potential field algorithm, etc. The A* algorithm, as a heuristic search algorithm, has the advantages of simplicity and ease of use. However, in the actual planning and design, it is found that the planned path has many problems, such as a large number of redundant nodes, a long search time, and an inaccurate and smooth path.

Researchers have made numerous innovations and improvements in this regard. Literature [1] improved the traditional A* algorithm by using the jump-point search method, which can significantly reduce the scale of useless nodes. But the disadvantage is that the designed path is not smooth enough when changing direction, and excessive sharp turns at large angles are not conducive to the driving and safety of the mobile robot. Literature [2] improved and adjusted the search neighborhood of the traditional A* algorithm, increasing the number of search neighborhoods to 24, thereby increasing the number of search angles [3] . However, it also brought about the problem of direct path finding through obstacles and inaccurate route planning. Literature [4] uses Bidirectional search to improve the A* algorithm. A* search is performed simultaneously at the starting point and the target point to get the same path at double speed and half the time. However, this improvement also has disadvantages. In a complex obstacle environment, searching in different directions will bring about the superposition of search areas, which will increase the time of judgment and processing and the difficulty of optimization.

Based on the aforementioned research, it is evident that the A* algorithm still faces several challenges. These challenges include the presence of excessive redundant nodes during expansion, slow search times, and the generation of sharp turning paths with large angles during planning. In response to these issues, a new approach is proposed: a five-neighborhood search A* algorithm that incorporates dynamic weighting in the heuristic function and utilizes second-order Bezier curve for path smoothing.

II. Traditional A* algorithm

The core idea of the traditional A* algorithm is to find the minimum-cost path from the starting point to the target point within a constrained map. The cost is determined by both the actual cost and the estimated cost. For nodes located at different positions, the cost satisfies the movement cost estimation function $F(n)$:

$$F(n) = G(n) + H(n) \tag{1}$$

Where: n represents the current position node during the path search process, $G(n)$ denotes the actual cost from the current position to the starting point, $H(n)$ represents the estimated cost from the current position to the target position, also known as the heuristic function.

The performance of the A* algorithm depends on the choice of the heuristic function H(n). Common heuristic functions include Manhattan distance, Euclidean distance, and diagonal distance [5]. Assuming the coordinates of the starting point are (x_n, y_n) and the coordinates of the goal point are (x_{goal}, y_{goal}).

The heuristic function for the Euclidean distance is:

$$H(n) = \sqrt{(x_n - x_{goal})^2 + (y_n - y_{goal})^2} \tag{2}$$

The heuristic function for the Manhattan distance is:

$$H(n) = |x_n - x_{goal}| + |y_n - y_{goal}| \tag{3}$$

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The heuristic function for the diagonal distance is:

$$H(n) = \sqrt{2} \times Diagonal + (Straight - 2 \times Diagonal) \quad (4)$$

Where: $Diagonal = \min(|x_n - x_{goal}| + |y_n - y_{goal}|)$, $Straight = |x_n - x_{goal}| + |y_n - y_{goal}|$

The Manhattan distance and diagonal distance have lower calculation accuracy than the Euclidean distance under the same conditions. In this article, the Euclidean distance is chosen as the heuristic function to maximize the possibility of obtaining the optimal path.

III. Improvements to the A* Algorithm

A. Weight Coefficient Improvement

In the traditional A* algorithm, the ratio between the estimated cost and the actual cost remains fixed, and both factors cannot be dynamically adjusted and insufficient bias during the path planning process. In the early to mid-stages of path exploration, the emphasis is on finding the shortest path and time, prioritizing search efficiency. During the later stages of path exploration, the focus shifts to achieving precision in reaching the target destination, and when approaching the goal, the path optimality becomes more important [6]; Hence, it is possible to improve the movement cost estimation function $F(n)$, resulting in improved search efficiency during the early and middle stages of the path exploration and increased accuracy in the later stages of the search. The improved $F(n)$ is:

$$F(n) = G(n) + W \times H(n) \quad (5)$$

By altering the value of W, the search efficiency is affected. Simulation comparisons reveal that different values of W lead to distinct search conditions, as shown in Table I. As the distance from the target position increases, the value of W should be set larger to achieve a quicker arrival at the selected location. Conversely, as the distance to the target position decreases, the value of W should be set smaller to ensure the accurate identification of the target point. Therefore, it is necessary to flexibly adjust the W value [7] in practical application to improve the efficiency of A* algorithm.

B. Search Neighborhood Improvement

In the traditional A* algorithm, the parent node needs to expand in eight directions to find the child node with the minimum cost, as shown in Fig. 1. In the absence of obstacles, the mobile robot can move in all eight directions around it. However, in the presence of obstacles and with specific environments defined for the starting and target points, as shown in Fig. 2. Searching through all eight directions will generate numerous redundant nodes, consuming a significant amount of search time and resulting in low efficiency.Therefore, it is necessary to make comprehensive judgments based on the starting point, target point, and obstacle environment, and flexibly improve the eight-search neighborhood to a five-search neighborhood.

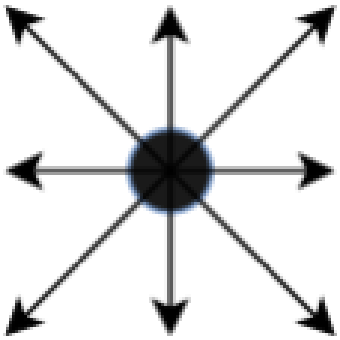


Fig. 1. Traditional A* Algorithm Eight Direction Search Graph



Fig. 2. Multi-directional Obstacle Blocking Diagram

Consider a scenario where the starting point and target point are positioned at the upper left and lower right, respectively, of the angle bisector of the second and fourth quadrants. Carry out traditional eight-direction expansion calibration on the starting point and the target point, and set the expansion direction from the starting point to the target point in the form of vector as shown in Fig. 3. It is known that the optimal vector expansion range is $[270° \sim 360°]$. By searching within this range, the mobile robot can reach the target point faster and more efficiently from the starting point.The traditional eight-direction expansion idea will extend the expansion range to $(0° \sim 270°)$ area. However, the expansion range $(90° \sim 180°)$ is contrary to the set optimal vector expansion range.In the early and middle stages of path planning, more attention paid to the shortest path and time.The expansion of the range of $(90° \sim 180°)$ will aggravate the path length and thus improve the path-finding time, so it is necessary to limit the expansion of this range.When the range of $[0° \sim 90°]$ and $[180° \sim 270°]$ are expanded at the same time, only by expanding in the range of $[0° \sim 90°]$ can all the expanded minimum cost point sets conform to the characteristics of bypassing obstacles. Considering comprehensively, the extension range of the selected

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TABLE I
COMPARISON TABLE OF WEIGHT COEFFICIENTS

$H(n)$	w value	Weight allocation ratio	Number of search points	search speed	route	cost
Euclidean distance	$w = 1$	$G(n) = H(n)$	medium	medium	medium	medium
	$w > 1$	$G(n) < H(n)$	few	fast	Non optimal	tall
	$0 < w < 1$	$G(n) > H(n)$	many	slow	optimal	low

child node is $[0^\circ \sim 90^\circ] \cup [270^\circ \sim 360^\circ]$, which is five of the eight traditional directions.Improve the traditional eight-neighborhood search, abandon three directions and change to five directions, as shown in Fig. 4. It can improve the search efficiency, greatly reduce the number of unnecessary redundant nodes, and save a lot of search time.

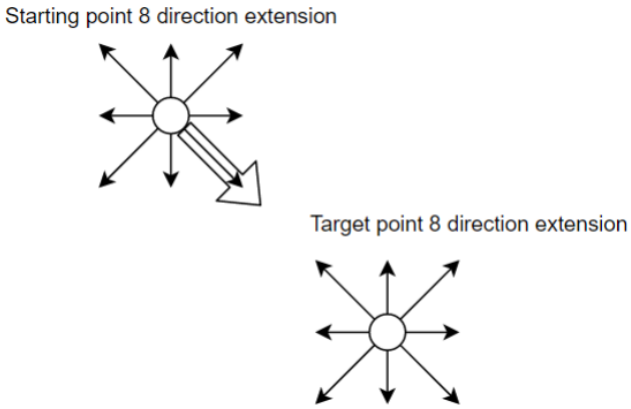


Fig. 3. Extension Vector Diagram from Starting Point to Target Point

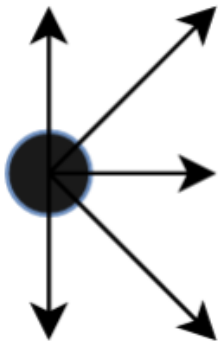


Fig. 4. Five-direction Search Diagram

C. Bezier curve optimization

The path generated by the improved A* algorithm has many problems, such as many broken lines, unsmooth trajectory and large corner of some paths, which will cause the robot to spend more time when driving along this path, and the battery power carried by the robot is limited, which will also reduce the effective working time of the robot. Therefore, in response to this issue, this article proposes a method of integrating Bezier curves to optimize the trajectory.For the robot working in the environment of this article, Path optimization by fusing second-order Bessel curves can meet the requirements in terms of smoothness. The formula for the second-order Bessel curve [8] is as follows:

$$B(t) = (1-t)^2 P_0 + 2t(1-t)P_1 + t^2 P_2, \quad t \in [0,1] \quad (6)$$

The principle is briefly described as follows: select three points in the plane and connect them in turn, Then, point G and point H are taken on the line segments DE and EF respectively, and DG/GE=EH/HF is satisfied.As shown in Fig. 5, connect GH again, take a point M on the line segment GH and the point M should satisfy DG/GE=EH/HF=GM/MH. As shown in Fig. 6, the second-order Bezier curve can be obtained by finding all the points M that meet the above conditions.

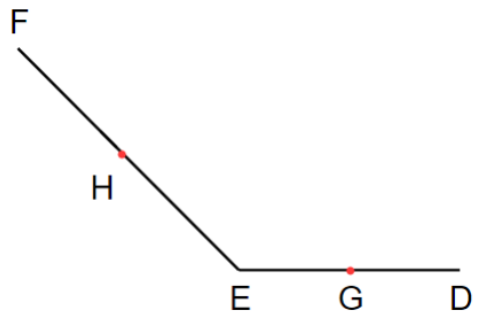


Fig. 5. Intermediate Point Selection Diagram

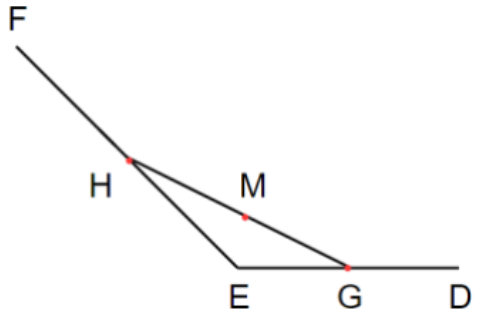


Fig. 6. Path points after optimization

IV. SIMULATION RESULTS AND ANALYSIS

To prove the effectiveness of the above improvements, a complex obstacle simulation map was designed in PyCharm.

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Simulation and comparative verification showed that the A* algorithm, after dynamic weight improvement, five neighborhood improvement, and Bessel curve path smoothing, has the characteristics of precise path finding, fewer irrelevant redundant nodes, fast search time, smooth path, and optimal cost.

A. Five Neighborhood Weighted Improvement Simulation Experiment

Design a simulation map with a size of 140 × 140, as shown in Fig. 7. Set the dot on the upper left side of the map as the starting point, the fork point on the lower right side as the target point, the grid size and the radius of the robot chassis are 2 and 1, respectively. Many black points form obstacles, The white area is regarded as a programmable path area. the green fork is the node traversed, and the routes before and after Bezier curve smoothing are red and blue respectively.

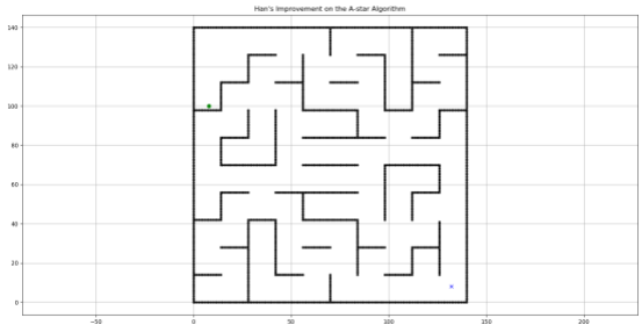


Fig. 7. 140*140 Simulation Map

Under the condition of having the same simulation map with the same starting point, target point, obstacle size and position, the simulation experiments of traditional eight-neighborhood A* search, eight-neighborhood dynamic weighted A* search and five-neighborhood dynamic weighted A* search are carried out respectively. The obtained simulation results are shown in Fig. 8, Fig. 9 and Fig. 10.

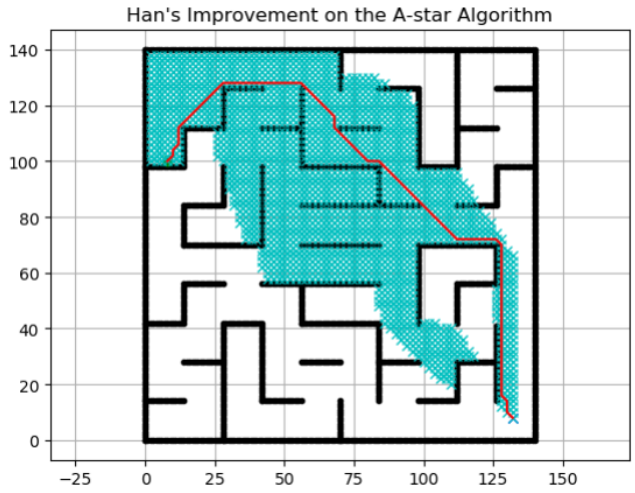


Fig. 8. Simulation Diagram of Traditional Eight Neighborhood Path Planning

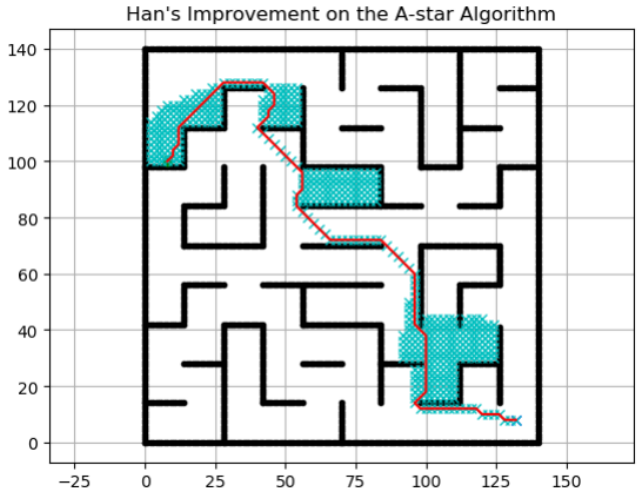


Fig. 9. Simulation Diagram of Improved Eight Neighborhood Dynamic Weighted Path Planning

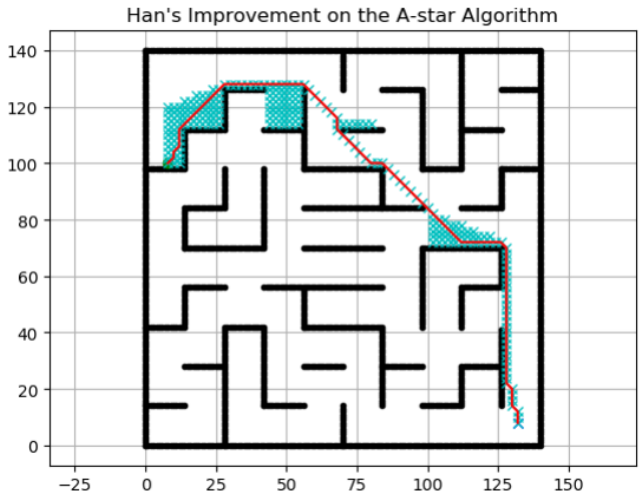


Fig. 10. Simulation Diagram of Improved Five Neighborhood Dynamic Weighted Path Planning

From Fig. 8 to Fig. 10, it can be seen that under the same map environment test, the A* algorithm after dynamic weighting of five neighborhoods can significantly reduce the range and number of redundant nodes searched, also improve the efficiency. The number of redundant nodes, search time and cost data obtained under the statistical simulation test are sorted out to get Table II.

From the data in the table, it can be concluded that the number of redundant nodes and the search time of the A* algorithm after dynamic weighting of five neighborhoods are reduced by 86.57% and 82.07% compared with the traditional eight-neighborhood search algorithm, and the cost remains the same in the precision of ten thousand bits.

B. Simulation Test of Path Smoothing Processing

The path obtained by the five-neighborhood dynamic weighted A* algorithm is not optimal and has the problem of

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TABLE II
ALGORITHM PERFORMANCE COMPARISON TABLE

Environment map size	Algorithm	Number of redundant nodes	Search time	cost
120×120	Traditional Eight Neighborhood A* Algorithm	1757	18.07134	113.15432
	Improved A* Algorithm with Eight Neighborhood Dynamic Weighting	483	4.48411	122.81118
	Improved A* Algorithm with Dynamic Weighting of Five Neighborhood	236	3.05042	113.15432

sharp turn at large Angle, so the second-order Bessel curve is used for processing. Divide the minimum node set on the path into 7 groups for grouping optimization and fusion processing. The overall optimized path is shown in Fig. 11, and the local 102 to 127 corner area is enlarged as shown in Fig. 12. From the overall and local graphs, it can be seen that the path obtained after Bezier curve processing has no large angle turning points. During the driving process, the robot is usually stuck due to sharp turning problems such as right angles and sharp corners. After smooth optimization, the continuity of speed and acceleration can be maintained during the driving process, and the driving fluency can be improved.

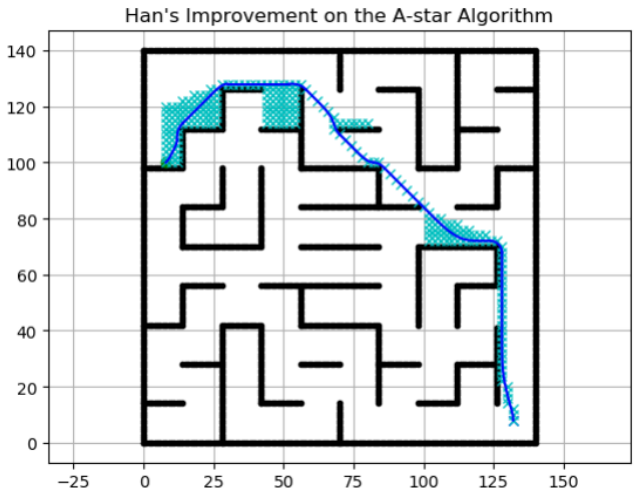


Fig. 11. Simulation Diagram of Whole Path Smoothing Processing

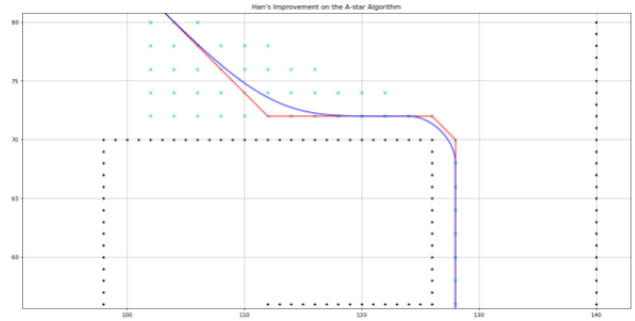


Fig. 12. Enlarged view of local corner areas from 102 to 127

V. CONCLUSION

This article proposes an improved A* algorithm for path planning of mobile robots, which dynamically adjusts the proportion of estimated costs to actual costs through different distances, reducing the search time in the early and middle stages, also enhancing the accuracy in the later stages.Using the five neighborhood search method, the opposite phase region points are discarded through vector method, while following the principle of both vector direction priority and expanding the minimum cost point avoid obstacles. The directionality is discarded according to the angle range, thereby reducing the number of redundant nodes in the search.Smoothing optimization using second-order Bessel curves reduces the risk of large angle turns for mobile robots during driving.Through the comparison between the simulation results and the data in the data table, it can be seen that the A* algorithm after dynamic weighting of five neighborhoods, combined with second-order Bezier curve smoothing optimization, is reasonable and effective in the path planning direction of mobile robots.

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